Chapter test

21 Photosynthesis

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. Which of the following statements about photosynthesis is ***incorrect***?

   A. It is a catabolic process.
   B. It is an energy conversion process.
   C. It is a process by which producers make food.
   D. It helps maintain the balance of oxygen and carbon dioxide in the atmosphere. ☐

2. The following word equation for photosynthesis is incomplete.

$$\text{X + water} \xrightarrow[\text{chlorophyll}]{\text{Y}} \text{glucose + Z}$$

   What do X, Y and Z represent, respectively?

	X	***Y***	***Z***
A.	oxygen	carbon dioxide	light
B.	oxygen	light	carbon dioxide
C.	light	carbon dioxide	oxygen
D.	carbon dioxide	light	oxygen ☐

★ 3. Which of the following is ***not*** an adaptation of leaves to carry out photosynthesis?

   A. Leaves are thin.
   B. The cuticle covering the leaf surfaces is transparent.
   C. The cells in the palisade mesophyll are tightly packed.
   D. The cells in the spongy mesophyll contain fewer chloroplasts. ☐

4. Before testing a leaf for starch it is put in hot alcohol. What is the purpose of this step?

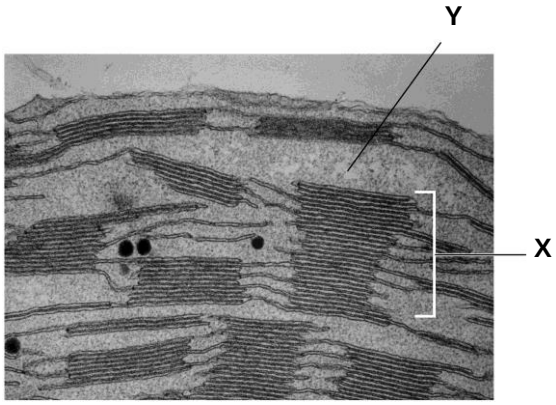
   A. to kill the cells of the leaf
   B. to stop the enzymatic reactions in the cells
   C. to remove the cuticle on the leaf surfaces
   D. to remove chlorophyll from the leaf ☐

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5. In photosynthesis, light is directly involved in

A. the synthesis of chlorophyll.
B. the activation of chlorophyll.
C. the reduction of NADP.
D. the formation of glucose. ☐

Directions: Questions 6 and 7 refer to the photomicrograph below, which shows part of a chloroplast.



6. Which is structure X?

A. stroma
B. thylakoid
C. granum
D. starch granule ☐

7. Which is a function of Y?

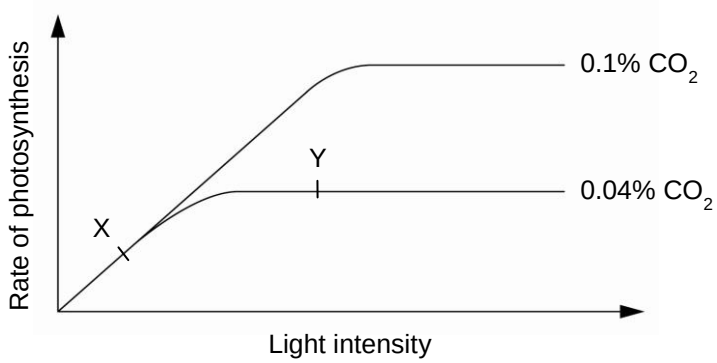
A. absorption of light energy
B. carbon fixation
C. production of ATP
D. storage of glucose ☐

8. What chemical substances are required by plants to synthesize proteins?

	Chemicals synthesized by the plant	***Chemicals absorbed from the soil***
A.	carbohydrates	magnesium compounds
B.	carbohydrates	nitrates
C.	vitamins	magnesium compounds
D.	vitamins	nitrates ☐

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9. The graph below shows the effects of light intensity and carbon dioxide concentration on the rate of photosynthesis.

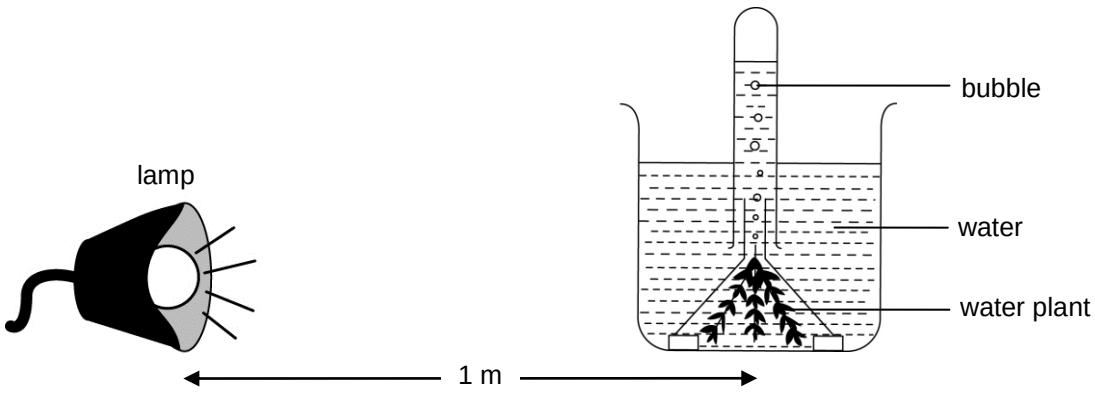


What limits the rate of photosynthesis at points X and Y respectively?

	X	***Y***
A.	light intensity	carbon dioxide concentration
B.	light intensity	temperature
C.	carbon dioxide concentration	light intensity
D.	temperature	carbon dioxide concentration

☐

★ 10. The diagram below shows a set-up used to investigate photosynthesis of a water plant.



Which of the following modifications of the set-up would increase the rate of bubble formation from the water plant?

(1) moving the table lamp 0.5 m closer to the water plant
(2) increasing the temperature of the water to 80 ˚C
(3) replacing the water with hydrogencarbonate solution

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

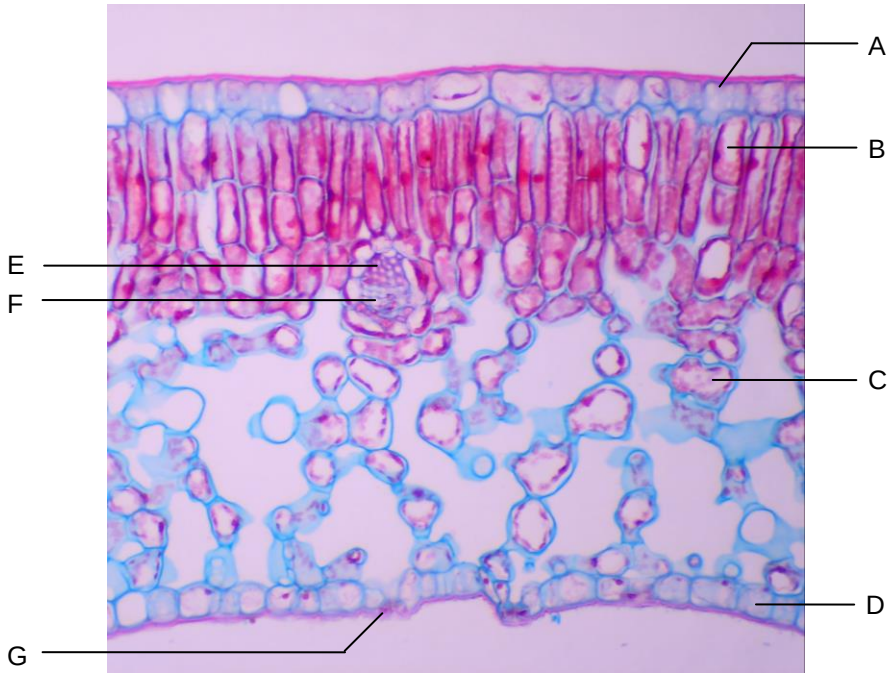
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Question	1	2	3	4	5	6	7	8	9	10
Related section	21.1	21.1	21.3	21.2	21.4	21.3	21.4	21.5	21.6	21.6

B. Problem solving (23 marks)

★ 11. The photomicrograph below shows the transverse section of a dicotyledonous leaf.



(a) Name cell types C and G. (2 marks)

(b) Cell type B has the highest photosynthetic rate. Describe ***two*** structural features which enable it to carry out photosynthesis effectively. (2 marks)

(c) Describe how cell type B obtains

(i) water from cell type E. (2 marks)

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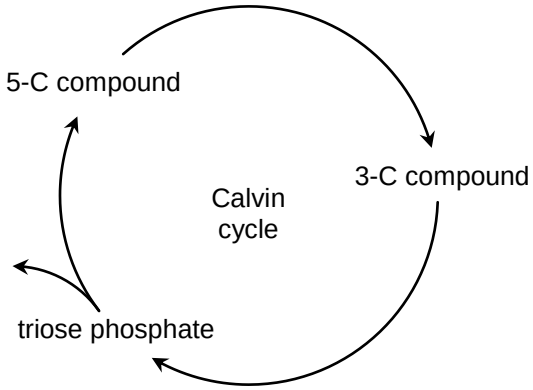
(ii) carbon dioxide from the atmosphere. (3 marks)

(d) (i) State the carbohydrate stored in cell type B. (1 mark)

(ii) If you are provided with a thin section of a leaf, how would you show the presence of the stored carbohydrate in it? (3 marks)

(Total: 13 marks)

★ 12. The diagram below shows an outline of the Calvin cycle.



(a) Complete the diagram to show where carbon dioxide enters the Calvin cycle. (1 mark)

(b) Name the reducing agent required by the Calvin cycle, and state in which conversion process of the Calvin cycle it is used. (2 marks)

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(c) Only some of the triose phosphate is used to regenerate the 5-C compound. The remaining triose phosphate may be used to synthesize carbohydrates. Suggest ***two*** other types of molecules that can be synthesized using triose phosphate. (2 marks)

(d) The reactions in the Calvin cycle are controlled by enzymes.

(i) State precisely where in a cell these enzymes are located. (1 mark)

(ii) In an investigation, the rate of photosynthesis of a plant was measured at different temperatures. The results are shown below.

Temperature (°C)	**Rate of photosynthesis (arbitrary units)**
5	2.2
10	4.1
15	4.5
20	4.6
25	2.5
30	0.0

A general statement is that

'the rate of a chemical reaction doubles for every 10 °C rise in temperature'.

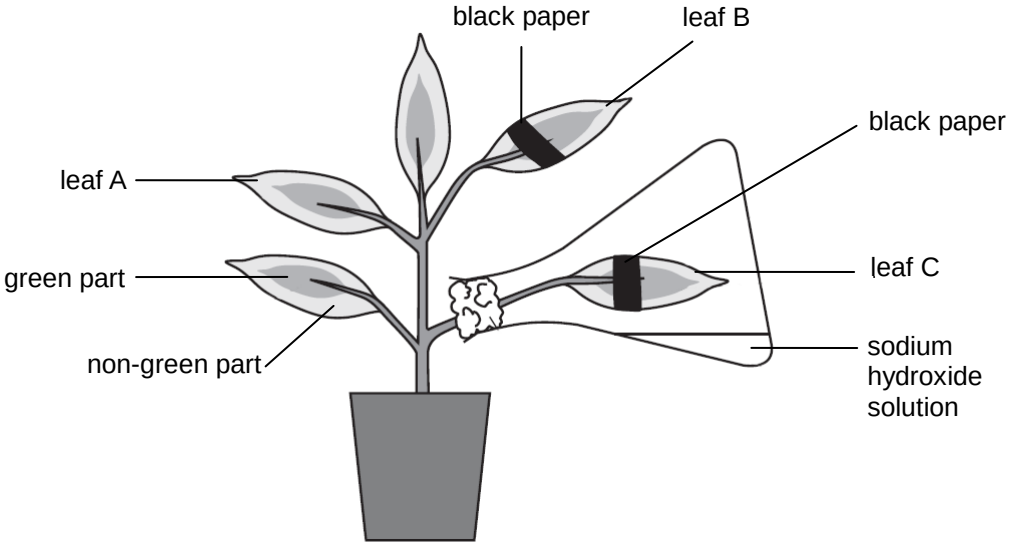
To what extent do the results of this investigation confirm the statement? Explain any differences observed. (4 marks)

(Total: 10 marks)

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C. Exam challenge (7 marks)

★★ 13. To study the conditions required for photosynthesis a destarched plant with variegated leaves was put under the sunlight for four hours as shown in the diagram below.



(a) Explain why the plant can be destarched by keeping it in darkness for 48 hours. (2 marks)

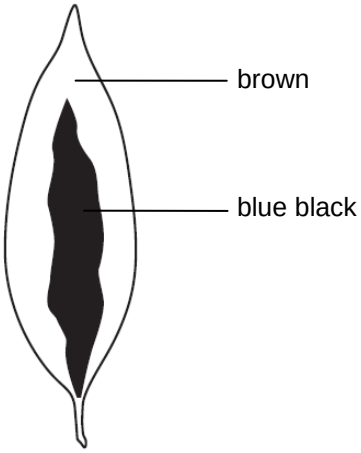
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_____

(b) What is the use of sodium hydroxide solution in the set-up? (1 mark)

_____

(c) After four hours, leaves A, B and C were detached and tested with iodine solution. The diagram below shows the appearance of leaf A after the test.



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In the space provided, draw labelled diagrams to show the appearance of leaf B and leaf C after the test. (2 marks)

(d) What conclusion, if any, can be drawn by comparing the results of

(i) the green part of leaf A and the green part of leaf B covered with black paper. (1 mark)

(ii) the green part of leaf A and the non-green part of leaf C. (1 mark)

(Total: 7 marks)

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Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	A	2.	D	3.	D	4.	D	5.	B
6.	C	7.	B	8.	B	9.	A	10.	C

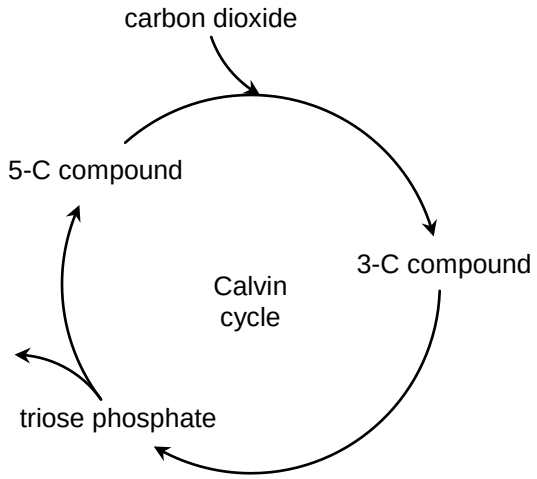
(1 mark each)

B. Problem solving (23 marks)

11. (a) C: spongy mesophyll cell (1)
G: guard cell (1)
(b) The cells are closely packed (1)
and contain a large number of chloroplasts to absorb more light. (1)
(c) (i) Water moves from cell to cell down the water potential gradient (1)
by osmosis. (1)
(ii) Carbon dioxide in the air diffuses through stomata (1)
into the air spaces in the leaf. (1)
Carbon dioxide dissolves in the water film on the surfaces of cell type B and then diffuses into the cells. (1)
(d) (i) starch (1)
(ii) Add a drop of iodine solution to the leaf section. (1)
Observe the leaf section under the light microscope. (1)
The section will turn blue black if starch is present. (1)

(Total: 13 marks)

12. (a)



(1)

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(b) NADPH (1)

It is used in the regeneration of the 5-C compound from triose phosphate. (1)

(c) Proteins (1)

Lipids (1)

(d) (i) stroma of a chloroplast (1)

(ii) The statement applies when the temperature increases from 5 °C to 15 °C that the rate of photosynthesis doubles. (1)

The increase in the rate of photosynthesis is small when temperature increases from 10 °C to 20 °C. (1)

The rate of photosynthesis decreases when temperature increases from 15 °C to 25 °C / from 20 °C to 30 °C. (1)

The optimum temperature is between 20 °C to 25 °C. / When the temperature is at 30 °C or above, the enzymes involved in Calvin cycle are denatured. (1)

(Total: 10 marks)

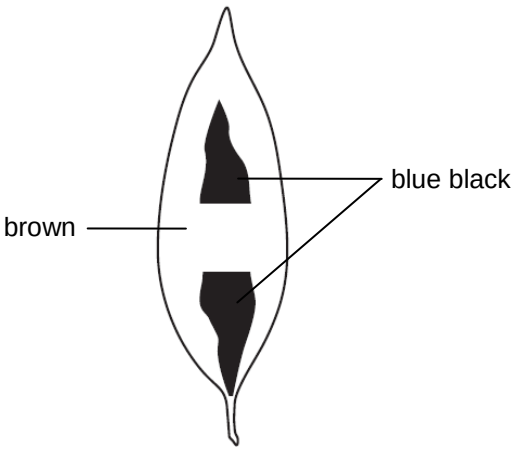
C. Exam challenge (7 marks)

13. (a) In darkness, starch in the leaves is converted to sugars (1)

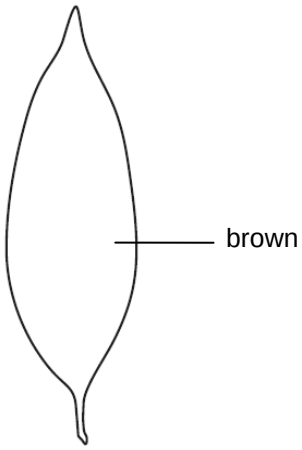
which are transported away from the leaves / oxidized in respiration. (1)

(b) To absorb carbon dioxide in the flask (1)

(c)



Appearance of leaf B after iodine test



Appearance of leaf C after iodine test

(2)

(d) (i) Light is necessary for photosynthesis. (1)

(ii) No conclusion can be drawn. (1)

(Total: 7 marks)

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